

DD-Mamba: Dual-Domain State-Space Forecasting with Linear Anchors and Zero-Initialized Corrections

Anonymous Submission

ADMA 2026 double-blind review

Abstract. Long-term multivariate time series forecasting must reconcile two complementary views of a signal: local, trend-driven dynamics in the time domain, and global periodic structure most compact in the frequency domain. Existing state-space (Mamba) forecasters model one view but not both, and deep forecasters often discard the simple linear maps that remain near state-of-the-art on several benchmarks. We present DD-Mamba, a dual-domain architecture whose time branch (trend–seasonal decomposition with a selective state-space encoder) and frequency branch (learnable complex spectral filtering) each produce a forecast, fused by a learned convex gate that structurally preserves both prediction paths. Two principles drive accuracy: (i) *linear anchoring* — the time branch carries a DLinear-style linear pathway, the frequency branch an optional FITS-style spectral map, and all nonlinear heads are zero-initialized, so at initialization the model reduces to a linear map of the instance-normalized window; and (ii) *capacity placement* — a bidirectional Mamba over the *variate* dimension carries the model on high-channel data and is disabled on small, few-channel data where it only overfits. On nine benchmarks under the look-back-96 protocol (three seeds), DD-Mamba reports the lowest mean MSE on seven, improving over the reported S-Mamba results on five (by up to 17% on Solar), tying three within run-to-run noise, and trailing one, with far fewer parameters on small datasets. Two-dataset ablations show the variate mixer as the one consistently validated component, while the frequency branch, temporal Mamba, and instance normalization are dataset-dependent — the latter helps weather and traffic yet hurts solar — so components must be validated per dataset.

Keywords: Time series forecasting · State-space models · Mamba · Frequency domain · Multivariate analysis

1 Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [1–3], patch-based attention

[4], inverted variate-token attention [5], and, most recently, selective state-space models (SSMs) built on Mamba [11, 12], which offer linear-time sequence modeling with input-dependent dynamics.

Two empirical facts complicate this progress. First, *simple linear maps remain stubbornly competitive*: a per-component linear projection from the look-back window to the horizon (DLinear [6]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [10]) match or beat far larger models on several benchmarks, especially the ETT family. Deep architectures that ignore this fact spend their early training re-discovering a linear map, and on small datasets often never do. Second, *the useful inductive bias differs by domain*: local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain. Models committed to a single view — attention or SSMs over time steps only, or spectral mixing only [9] — systematically underuse the other.

We address both issues with DD-Mamba, a dual-domain forecaster (Fig. 1) organized around three ideas.

Dual branches with structurally preserved paths. A time branch and a frequency branch each output a complete forecast; the model’s prediction is a learned *convex combination* of the two, with the gate conditioned on both branches’ features. The output is constrained elementwise between the branch forecasts, so both domains retain an explicit prediction path, and the gate value reports the convex weight assigned to each branch — a proxy for, not a causal measure of, their contributions — while the *actual* contribution of each branch is learned, and the gate may concentrate almost entirely on one branch for some samples or datasets (it is a sigmoid and can saturate near 0 or 1).

Linear anchors with zero-initialized corrections. Each branch carries an explicit *linear* pathway that parameterizes a strong linear forecaster family of its domain: DLinear-style seasonal/trend window-to-horizon projections [6] in the time branch, and an optional FITS-style complex-linear spectral map [10] in the frequency branch. These are parameterized pathways trained jointly with the model — not pretrained or fitted linear predictors. All nonlinear heads (and the spectral map) are zero-initialized and the fusion gate opens as a constant g_0 , so at initialization the model reduces to g_0 times a randomly initialized DLinear-style map of the *instance-normalized* window — linear in the normalized space, the same normalization wrapper used by RLinear-class linear models [7, 8] (Sect. 3.2). This choice stabilizes small-data training and removes the mean-reversion failure mode of pooled deep forecasters.

Mamba where it pays. A causal Mamba encoder models the time axis; a *bidirectional* Mamba (frequency bins and channels have no arrow of time) serves two optional roles: spectral encoding, and cross-channel mixing over the variate dimension in the spirit of S-Mamba [12]. Crucially, capacity is *placed* per dataset: the variate mixer is the dominant component on high-channel data (electricity, traffic, solar) and is disabled on few-channel small data (ETT, exchange) where ablations show it only adds overfitting capacity.

Our contributions are:

1. A dual-domain forecasting architecture whose output is a gated convex combination of per-branch forecasts, structurally preserving a prediction path from each domain and exposing each branch’s per-sample convex weight (Sect. 3).
2. A linear-anchoring scheme — domain-appropriate linear pathways with zero-initialized nonlinear corrections — under which the model at initialization reduces to a linear map of the instance-normalized window, yielding large gains on small benchmarks (Sect. 3.2, ablated in Sect. 4.3).
3. A capacity-placement recipe for selective SSMS: causal Mamba over time where temporal microstructure matters, bidirectional Mamba over variates where cross-channel structure dominates, and neither where data is scarce (Sect. 3.4).
4. The lowest reported average MSE on seven of nine standard benchmarks under the unified look-back-96 protocol (three-seed mean $\pm$ std for every entry), relative to results published under the same protocol, plus a released ablation harness and multi-seed ablations on two datasets — including negative and dataset-dependent findings (instance normalization helps weather and traffic yet hurts solar) that we believe are as actionable as the positive ones (Sect. 4).

2 Related Work

Transformer forecasters. Informer [1], Autoformer [2] and FEDformer [3] adapt attention to long sequences via sparsity, decomposition, and frequency-domain mixing respectively. PatchTST [4] treats sub-series patches as tokens with channel independence; iTransformer [5] inverts the axes and attends over *variates*, establishing that cross-channel modeling, not temporal attention, drives accuracy on high-dimensional benchmarks. DD-Mamba adopts the variate-mixing insight but implements it with linear-time bidirectional SSMS and, unlike these works, couples it to an explicit frequency branch and linear anchors.

Linear and frequency-domain models. DLinear/NLinear [6] showed that decomposition plus a linear map rivals Transformers, particularly on ETT; RLinear [7] added reversible instance normalization [8]. FreTS [9] learns MLPs in the frequency domain, and FITS [10] forecasts with a single complex linear interpolation of the low-passed spectrum at negligible parameter cost. DD-Mamba does not compete with these models — it *contains* them: each branch carries a linear pathway of its domain’s model family, and the deep machinery is trained as a correction on top.

State-space models. Mamba [11] introduced selective SSMS with input-dependent dynamics and linear-time scans. In forecasting, S-Mamba [12] applies bidirectional Mamba over variate tokens with a linear temporal embedding, and further works explore frequency-domain SSMS [13]. DD-Mamba unifies these threads: causal Mamba over time, bidirectional Mamba over frequency bins and over variates, with the placement of each governed by dataset characteristics rather than applied uniformly.

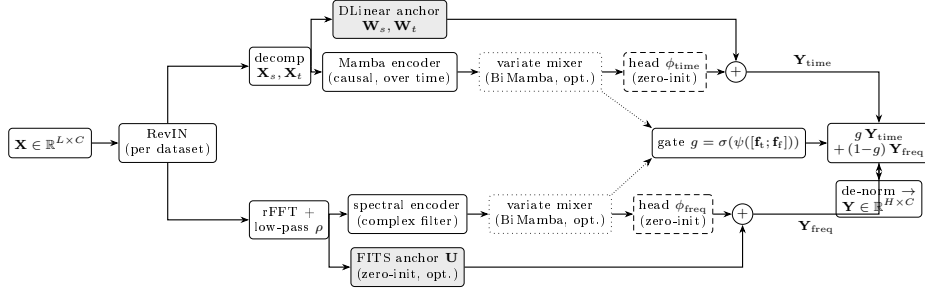


Fig. 1. DD-Mamba overview. Both branches carry a linear anchor (shaded) and zero-initialized nonlinear heads (dashed), so at initialization the model reduces to a linear map of the instance-normalized window. Dotted boxes are per-dataset switches; dotted arrows carry branch features to the fusion gate, whose convex weight g mixes the two branch forecasts.

3 Method

3.1 Problem Formulation and Normalization

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L \times C}$ of L steps and C variates, the task is to predict the next H steps $\mathbf{Y} \in \mathbb{R}^{H \times C}$. Following RevIN [8], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, and de-normalize the final forecast. Instance normalization is treated as a per-dataset switch rather than a universal default; Sect. 4.3 shows why.

3.2 Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [2, 6] and forecasts with two cooperating paths.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s \mathbf{X}_s + \mathbf{W}_t \mathbf{X}_t \in \mathbb{R}^{H \times C}, \quad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H \times L}, \quad (1)$$

i.e., a DLinear-style map [6] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [11]. A Mamba layer discretizes a state-space system with input-dependent parameters $(\boldsymbol{\Delta}, \mathbf{B}, \mathbf{C})$:

$$\mathbf{h}_k = e^{\boldsymbol{\Delta}_k \mathbf{A}} \mathbf{h}_{k-1} + \boldsymbol{\Delta}_k \mathbf{B}_k u_k, \quad y_k = \mathbf{C}_k^\top \mathbf{h}_k + D u_k, \quad (2)$$

scanned causally over the L steps and applied independently to each of the d_{in} feature dimensions: per dimension, the state is $\mathbf{h}_k \in \mathbb{R}^N$, $\mathbf{A} \in \mathbb{R}^N$ is diagonal, the input u_k is scalar, and $(\boldsymbol{\Delta}_k \in \mathbb{R}, \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^N)$ are produced from u by learned linear maps [11]. The last state, summed with a whole-window linear embedding

$\mathbf{W}_e \tilde{\mathbf{x}}^{(c)}$ (one per variate, as in [12]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C \times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \quad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \quad (3)$$

Because ϕ_{time} is zero-initialized, the branch’s output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express. We compute the recurrence of Eq. (2) in fp32 even under mixed-precision training: the exponential and the L -step state accumulation overflow in half precision, which silently destroys training (observed as NaN validation losses that exhaust early-stopping patience).

3.3 Frequency Branch: Spectral Filtering and a FITS Anchor

The frequency branch applies the real FFT along time, $\mathbf{Z} = \text{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C \times F}$, $F = \lfloor L/2 \rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i$ densely mixes bins, $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top \in \mathbb{C}^{C \times F}$ with $\mathbf{W} \in \mathbb{C}^{F \times F}$ acting along the frequency axis, realized with two real matrices; a bidirectional Mamba over the bin sequence is available as an input-dependent alternative. The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{C \times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS anchor (optional, per dataset). Mirroring the time branch, a complex linear map $\mathbf{U} \in \mathbb{C}^{F' \times F}$ with $F' = \lfloor (L+H)/2 \rfloor + 1$, applied along the frequency axis ($\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C \times F'}$), maps the (low-passed) window spectrum to that of a length- $(L+H)$ series, whose inverse transform’s last H samples are the branch’s linear spectral forecast:

$$\mathbf{Y}_{\text{freq}} = \underbrace{\text{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[L:]}_{\text{linear-in-frequency}} + \phi_{\text{freq}}(\mathbf{f}_{\text{freq}}), \quad \mathbf{U}, \phi_{\text{freq}} \equiv \mathbf{0} \text{ at init.} \quad (4)$$

This pathway *parameterizes* the FITS [10] family — any linear-in-frequency forecaster of the low-passed spectrum is reachable — but, unlike FITS itself, $\mathbf{U}$ is *zero-initialized*, so the frequency branch is silent at initialization rather than a functioning spectral predictor. This is deliberate: the frequency grids of a length- L and a length- $(L+H)$ window do not align (harmonic k of the input maps to the non-integer index $k(L+H)/L$ of the output), so no canonical identity or interpolation initialization exists; a randomly initialized $\mathbf{U}$ would inject noise into the fused forecast during the highest-learning-rate epochs. Zero initialization instead lets the gate lean on the time anchor until $\mathbf{U}$ is learned jointly with the rest of the model. Consequently, the precise initialization claim is: the time branch coincides with a (randomly initialized) DLinear, the frequency branch outputs zero, and the fusion gate is the constant g_0 (Sect. 3.5), so in the instance-normalized space the end-to-end map is the linear function $g_0 \cdot \mathbf{Y}_{\text{lin}}$. Composed with RevIN — whose window statistics are themselves a nonlinear function of the input — the model at initialization is a randomly initialized linear

map wrapped in instance normalization, exactly the structure of RLinear-class models [7, 8]; it is linear in the normalized window, not in the raw input, and it is linear by construction, not a fitted linear model. The anchor is enabled per dataset (ETTh1, ETTh2, and Weather; Table 2); where it is disabled, the frequency branch remains fully active through its spectral encoder and head but carries no explicit linear pathway, so the linear-anchoring property then rests on the time branch alone.

3.4 Cross-Channel Mixing over Variates

Channel independence is a strong baseline [4, 6] but leaves accuracy on the table when variates are strongly coupled [5, 12]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the variate dimension — treating the C per-channel features as a sequence — followed by a position-wise feed-forward block. Bidirectionality is essential: channel order carries no causal direction, so each layer runs forward and reversed scans and sums them in one residual stream. The mixer is a per-dataset choice: $M=2$ at width 512 on electricity/traffic (where it is the single most load-bearing component; Sect. 4.3), $M=2$ at width 256 on weather, $M=1$ on solar, and $M=0$ on the 7–8-channel ETT and exchange datasets, where it triples parameters without adding signal.

3.5 Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1-g) \odot \mathbf{Y}_{\text{freq}}, \quad g = \sigma(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])) \in (0, 1)^{C \times 1}, \quad \psi \text{ applied row-wise (per variate)}, \quad (5)$$

followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, both domains retain an explicit prediction path, and the gate value reports each branch’s convex mixing weight — a lightweight proxy for its contribution, not a causal attribution. The constraint is structural, not behavioral: g is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little for some samples or datasets — which is by design, as the ablations show the useful domain is dataset-dependent. ψ is zero-initialized with bias 2.2, so training starts at $g \approx 0.9$: the time branch’s output is linear (in the normalized window) at initialization while the frequency branch is silent, and an even 0.5/0.5 mix would waste the highest-learning-rate epochs compensating for noise.

3.6 Training

We minimize MSE with AdamW under a halving schedule ($\eta_e = \eta_0 \cdot 2^{-(e-1)}$), which concentrates learning in the first epochs and suppresses the late-epoch overfitting that cosine schedules invite on small benchmarks. All runs use 10 epochs; early stopping monitors validation loss with the best checkpoint restored.

Table 1. Benchmark datasets and where DD-Mamba places its capacity.

Dataset	Variates	Steps	Granularity	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	off	0.25M
ETTm1/ETTm2	7	69,680	15 min	off	0.24–0.36M
Weather	21	52,696	10 min	$M=2$, $d=256$	5.8M
Solar-Energy	137	52,560	10 min	$M=1$, $d=128$	0.96M
Electricity	321	26,304	1 hour	$M=2$, $d=512$	19.7M
Traffic	862	17,544	1 hour	$M=2$, $d=512$	19.7M
Exchange	8	7,588	1 day	off	0.09M

4 Experiments

4.1 Setup

Datasets and protocol. We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [5, 12]: look-back $L=96$ for all datasets and horizons $H \in \{96, 192, 336, 720\}$; ETT uses the canonical 12/4/4-month border split (the file tail beyond month 20 is unused), the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE on the standardized test set.

Baselines. We compare against the strongest published results under the same protocol: the state-space forecaster S-Mamba [12], iTransformer [5], PatchTST [4], DLinear [6], and TimesNet [14]. Baseline numbers are quoted from [12] rather than reproduced: the evaluation protocol (look-back, splits, standardization, metrics) is identical, but training pipelines, seeds, and implementations differ across works, so sub-noise-floor gaps to quoted numbers should be read with caution. We flag every such case in the text, and re-running all baselines within one unified training framework is planned for the camera-ready appendix.

Implementation and Reproducibility. All experiments use PyTorch 2.11 (CUDA 13.0) on a single NVIDIA RTX 5090 (32 GB). Optimization is AdamW with gradient clipping at 5.0, 10 epochs with the learning rate halved after every epoch, early stopping on validation loss with the best checkpoint restored, and mixed precision with the selective scan forced to fp32 (Sect. 3.2). The decomposition kernel is 25; Mamba blocks use expansion 2 and depthwise convolution width 4 throughout. Table 2 lists all per-dataset settings; complete configuration files and the ablation harness ship with the code. All main results (Tables 3 and 4) are means over *three seeds* (2024–2026); Table 4 reports the per-entry standard deviations, whose maximum across all 36 dataset–horizon entries is 0.005 MSE (Solar, $H=720$) with a typical value of 0.001–0.003. This confirms the ≈ 0.005 noise floor used for the tie rule empirically. As a first unified-framework baseline, we re-trained DLinear through the identical data pipeline, splits, schedule, and evaluation script on ETTh1 (three seeds): it averages 0.462/0.456 MSE/MAE, within 0.006 of its quoted result (0.456/0.452), supporting the comparability

Table 2. Per-dataset hyperparameters. d : model width; time enc.: time-branch encoder (Mamba layers in parentheses); M : variate-mixer layers; N : SSM state size; ρ : spectral low-pass fraction (Sect. 3.3); FITS: the *optional* spectral linear anchor of Eq. (5) — when “no”, the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. [†]Solar adopts IN = no from the ablation finding (Sect. 4.3); its main results were re-measured under this configuration across all horizons and seeds.

Dataset	d	time enc.	M	N	ρ	FITS	IN	lr	B	do./wd
ETTh1	128	Mamba (1)	0	16	0.4	yes	yes	$5 \cdot 10^{-4}$	32	$0.2/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.3	yes	yes	$5 \cdot 10^{-4}$	32	$0.3/5 \cdot 10^{-4}$
ETTM1	128	Mamba (2)	0	16	0	no	yes	10^{-4}	32	$0.1/10^{-4}$
ETTM2	128	Mamba (1)	0	16	0.2	no	yes	10^{-4}	32	$0.2/2 \cdot 10^{-4}$
Weather	256	Mamba (2)	2	16	0	yes	yes	$2 \cdot 10^{-4}$	32	$0.1/10^{-4}$
Solar	128	Mamba (2)	1	16	0	no	no [†]	$5 \cdot 10^{-4}$	16	$0.1/10^{-4}$
Electricity	512	MLP	2	32	0	no	yes	$5 \cdot 10^{-4}$	16	$0.1/10^{-4}$
Traffic	512	MLP	2	32	0	no	yes	10^{-3}	16	$0.1/10^{-4}$
Exchange	64	Mamba (1)	0	8	0.5	no	yes	10^{-4}	32	$0.3/5 \cdot 10^{-4}$

Table 3. Long-term forecasting results averaged over horizons $H \in \{96, 192, 336, 720\}$ with look-back $L=96$ (MSE/MAE; lower is better). DD-Mamba entries are means over three seeds (Table 4 reports per-horizon std). Baselines quoted from [12]. **Bold** marks the best result *and* every result within the run-to-run noise floor (< 0.005 , Sect. 4.1) of it, treated as tied best; underline marks the next-best result.

Dataset	DD-Mamba (ours)	S-Mamba	iTransformer	PatchTST	DLinear	TimesNet
ETTh1	0.439/0.434	0.455/0.450	<u>0.454/0.447</u>	0.469/0.454	0.456/0.452	0.458/0.450
ETTh2	0.374/0.401	<u>0.381/0.405</u>	<u>0.383/0.407</u>	<u>0.387/0.407</u>	0.559/0.515	0.414/0.427
ETTM1	0.386/0.398	<u>0.398/0.405</u>	0.407/0.410	0.387/0.400	0.403/0.407	0.400/0.406
ETTM2	0.280/0.325	<u>0.288/0.332</u>	<u>0.288/0.332</u>	0.281/0.326	0.350/0.401	0.291/0.333
Weather	0.248/0.276	0.251/0.276	<u>0.258/0.278</u>	<u>0.259/0.281</u>	0.265/0.317	0.259/0.287
Electricity	0.168/0.263	0.170/0.265	<u>0.178/0.270</u>	0.205/0.290	0.212/0.300	0.192/0.295
Traffic	<u>0.438/0.281</u>	0.414/0.276	<u>0.428/0.282</u>	0.481/0.304	0.625/0.383	0.620/0.336
Exchange	<u>0.363/0.405</u>	0.367/0.408	<u>0.360/0.403</u>	<u>0.367/0.404</u>	0.354/0.414	0.416/0.443
Solar	0.199/0.260	0.240/ <u>0.273</u>	<u>0.233/0.262</u>	0.270/0.307	0.330/0.401	0.301/0.319

of quoted numbers under this protocol; DD-Mamba’s margin over it (0.439 vs. 0.462) is roughly six times the largest observed std. Re-running the remaining baselines in the same framework is left for the camera-ready appendix.

4.2 Main Results

Table 3 summarizes averages over the four horizons; Table 4 reports DD-Mamba per horizon with per-entry standard deviations. Applying the noise-floor tie rule consistently, DD-Mamba reports the lowest average MSE on seven of the nine datasets. The margin over the best baseline is clear on Solar (0.199 vs. iTransformer’s 0.233; -17% vs. S-Mamba, roughly seven times the largest observed std — measured with the instance-normalization-free configuration from Sect. 4.3), on ETTh1 (-0.015), and on ETTh2 (-0.007); on ETTM1/ETTM2 the margin over S-Mamba is clear ($-0.012/-0.008$) but the gap to PatchTST is 0.001 —

Table 4. DD-Mamba per-horizon results (mean $\pm$ std over three seeds), look-back $L=96$.

Dataset		$H=96$	$H=192$	$H=336$	$H=720$	Avg
ETTh1	MSE	0.380 $\pm$.001	0.431 $\pm$.002	0.474 $\pm$.004	0.470 $\pm$.003	0.439
	MAE	0.396 $\pm$.001	0.424 $\pm$.001	0.447 $\pm$.003	0.468 $\pm$.001	0.434
ETTh2	MSE	0.293 $\pm$.001	0.368 $\pm$.003	0.414 $\pm$.001	0.421 $\pm$.001	0.374
	MAE	0.344 $\pm$.002	0.391 $\pm$.000	0.426 $\pm$.000	0.441 $\pm$.001	0.401
ETTM1	MSE	0.320 $\pm$.001	0.366 $\pm$.002	0.398 $\pm$.001	0.462 $\pm$.001	0.386
	MAE	0.359 $\pm$.001	0.384 $\pm$.001	0.406 $\pm$.001	0.443 $\pm$.001	0.398
ETTM2	MSE	0.177 $\pm$.000	0.242 $\pm$.002	0.302 $\pm$.000	0.401 $\pm$.003	0.280
	MAE	0.260 $\pm$.001	0.303 $\pm$.001	0.340 $\pm$.000	0.398 $\pm$.002	0.325
Weather	MSE	0.162 $\pm$.000	0.212 $\pm$.000	0.269 $\pm$.001	0.352 $\pm$.003	0.248
	MAE	0.207 $\pm$.001	0.255 $\pm$.000	0.295 $\pm$.002	0.349 $\pm$.002	0.276
Electricity	MSE	0.140 $\pm$.000	0.157 $\pm$.001	0.174 $\pm$.001	0.200 $\pm$.004	0.168
	MAE	0.235 $\pm$.000	0.252 $\pm$.001	0.271 $\pm$.001	0.296 $\pm$.004	0.263
Traffic	MSE	0.402 $\pm$.002	0.424 $\pm$.001	0.441 $\pm$.001	0.485 $\pm$.003	0.438
	MAE	0.266 $\pm$.001	0.275 $\pm$.001	0.283 $\pm$.000	0.301 $\pm$.000	0.281
Exchange	MSE	0.086 $\pm$.000	0.179 $\pm$.001	0.330 $\pm$.002	0.857 $\pm$.001	0.363
	MAE	0.205 $\pm$.000	0.301 $\pm$.000	0.416 $\pm$.002	0.699 $\pm$.000	0.405
Solar	MSE	0.180 $\pm$.004	0.195 $\pm$.005	0.207 $\pm$.001	0.212 $\pm$.005	0.199
	MAE	0.237 $\pm$.003	0.259 $\pm$.003	0.270 $\pm$.001	0.274 $\pm$.005	0.260

a tie; on Weather (-0.003) and Electricity (-0.002) the lead over S-Mamba is below the floor and the methods should be read as tied. DD-Mamba remains competitive on Exchange — a near random-walk dataset where all methods cluster within 0.013 MSE and plain DLinear is best, 0.009 ahead of DD-Mamba — and lags S-Mamba on Traffic by $+0.024$ MSE. We *hypothesize* the traffic deficit stems from the weekly cycle exceeding the protocol look-back; this is not yet experimentally verified, and Sect. 5 lists the experiments that would test it. All comparisons are relative to results *reported* in [12] under the same protocol (Sect. 4.1), so we phrase our overall claim conservatively: DD-Mamba is competitive with results reported under the same protocol, with its clearest margins on the small ETT benchmarks. The clearest margins appear on the small ETT datasets, where the linear anchors and deliberate capacity restraint matter most; there DD-Mamba uses roughly 0.25M parameters, two orders of magnitude fewer than variate-attention models configured for those datasets.

4.3 Component Ablations: A Case Study

We release an ablation harness in which every variant flips exactly one switch of a dataset’s full configuration, holding the split, schedule, seeds, and all other hyperparameters fixed. We report its complete output on two datasets — Solar-Energy ($H=96$, single seed, Table 5) and Weather ($H=96$, *three seeds*, Table 6) — together with targeted probes on traffic and ETT that changed the released configurations. Deltas are interpreted against the empirical noise floor of Sect. 4.1; sub-floor deltas are reported as ties. Extending the multi-seed matrix to the remaining datasets and horizons is left for the camera-ready version.

Table 5 isolates each component on Solar-Energy. The “full” configuration here predates the IN = no finding below (it uses IN = yes, scoring 0.197); acting on that finding, the final Solar configuration disables instance normalization, and

Table 5. Component ablation on Solar-Energy ($H=96$, single seed; noise floor ≈ 0.005 MSE). Each variant flips exactly one switch of the full configuration.

Variant	MSE	Δ MSE	Component isolated
full	0.197	—	reference
w/o channel mixer	0.219	+0.022	cross-variate Mamba mixing
w/o linear backbone	0.209	+0.012	DLinear anchor (time)
freq branch only	0.209	+0.012	time branch removed
sum fusion	0.202	+0.006	learned gate $\rightarrow$ average
time branch only	0.198	+0.002	frequency branch removed
Mamba $\rightarrow$ MLP over time	0.197	+0.001	selective scan (time)
w/o instance norm	0.181	−0.016	RevIN

the main results (Tables 3–4) were re-measured under it across all horizons and seeds — its $H=96$ mean of $0.180 \pm .004$ matches this table’s w/o-instance-norm variant (0.181), closing the loop between ablation and main results.

The evidence is differentiated, and we read it as such. *Clear contributors* (well above the noise floor): the variate mixer (+0.022 when removed) — consistent with 137 strongly coupled photovoltaic plants — the DLinear anchor (+0.012), and the time branch as a whole (+0.012). *Marginal on this dataset*: the learned gate over plain averaging (+0.006, barely above the floor), the frequency branch (+0.002, below the floor), and the temporal Mamba versus an MLP (+0.001, a tie). *Harmful here*: instance normalization (−0.016 when removed). We therefore do not claim that every component is validated by this study: on solar, the frequency branch and the temporal SSM show no measurable benefit, and their value — like RevIN’s — is dataset-dependent (the frequency branch is structurally limited whenever the dominant period exceeds the look-back, as here and on weather/traffic; the temporal SSM is dispensable exactly where the variate mixer carries the signal). Multi-dataset, multi-seed evidence is required before component-level claims generalize, and the released harness is built to produce it.

A second complete run of the harness on Weather ($H=96$, three seeds; Table 6) provides exactly that evidence and sharpens the picture. Two components are clear contributors on weather: instance normalization ($+0.0161 \pm .004$ when removed — the *opposite* sign of solar, now established with multi-seed statistics) and the variate mixer ($+0.0104 \pm .002$, consistent with solar). The temporal Mamba, a tie on solar, shows a small but statistically significant benefit on weather ($+0.0027$, $>5\sigma$ given stds of .0005/.0003) — matching the analysis that weather’s accuracy rests on local temporal dynamics. The remaining components (single branches, FITS, gate vs. average, spectral encoder type) are within one–two std of the full model on weather. Across the two studies the consistent findings are: the variate mixer is the one component validated on both datasets; instance normalization is strongly dataset-dependent in *both* directions; and the value of the remaining components varies with dataset structure — which is the

Table 6. Component ablation on Weather ($H=96$, mean $\pm$ std over three seeds). Each variant flips exactly one switch of the full configuration.

Variant	MSE	Δ MSE (mean)	Component isolated
full	0.1620 $\pm$.000	—	reference
w/o instance norm	0.1781 $\pm$.004	+0.0161	RevIN
w/o channel mixer	0.1724 $\pm$.002	+0.0104	cross-variate Mamba mixing
Mamba $\rightarrow$ MLP over time	0.1647 $\pm$.001	+0.0027	selective scan (time)
w/o linear backbone	0.1631 $\pm$.001	+0.0011	DLinear anchor (time)
time branch only	0.1628 $\pm$.001	+0.0007	frequency branch removed
w/o FITS anchor	0.1627 $\pm$.002	+0.0007	spectral linear anchor
freq branch only	0.1626 $\pm$.001	+0.0006	time branch removed
freq Mamba encoder	0.1626 $\pm$.002	+0.0006	spectral encoder type
sum fusion	0.1622 $\pm$.001	+0.0001	learned gate $\rightarrow$ average

architectural argument for shipping them as per-dataset switches rather than universal defaults.

Normalization is a per-dataset decision. The same probe that revealed RevIN *hurting* solar (-0.016 MSE when removed) shows the opposite on traffic: removing instance normalization degrades traffic from 0.405 to 0.505 MSE at $H=96$. The two datasets look alike on the surface — both bounded, both strongly periodic — but solar’s night zeros are *periodic structure* (window statistics then mostly encode the window’s night fraction, and de-normalization amplifies errors by an unstable standard deviation), whereas traffic windows carry genuine distribution shift (weekday/weekend composition, per-sensor levels) that instance normalization absorbs. We therefore ship normalization as a verified per-dataset switch and caution against transferring normalization conclusions across datasets.

Negative results that shaped the design. Deepening the variate mixer on traffic from two to four layers (19.7M $\rightarrow$ 37.9M parameters) produced no accuracy gain, so the released configuration keeps two layers; our working *hypothesis* for the residual gap to S-Mamba on traffic is the weekly cycle, which at hourly sampling (168 steps) exceeds the protocol look-back of 96 and is invisible to both branches — a hypothesis directly testable by widening the look-back to 168/336, adding calendar covariates, or re-comparing against S-Mamba under a longer window, none of which we have run yet. Similarly, enabling the variate mixer on ETT (7 channels, 8.4K training windows) tripled parameters and *caused* premature early stopping through immediate overfitting; the released ETT configurations disable it.

4.4 Analysis

Where each domain wins. The branch-only ablations (Table 5) quantify the two domains directly on solar: removing the time branch costs $+0.012$ MSE while removing the frequency branch costs $+0.002$, i.e., both contribute but the time view carries more — consistent with the protocol look-back hiding part of the periodic structure (at 10-minute resolution the daily cycle spans 144 samples $> L=96$; the same argument applies to weather’s daily and traffic’s

weekly cycles). Beyond ablations, the fusion gate (Eq. 5) exposes per-sample, per-channel convex mixing weights at no extra cost — a proxy for, not a causal measure of, branch contributions; a systematic study of gate behavior across datasets is left for the extended version.

Efficiency. We make no measured efficiency claim in this version and note only two design properties: parameter counts span 0.09M–19.7M across datasets (Table 1), sized to each dataset rather than fixed; and the bidirectional variate mixer scans C tokens rather than $B \times C$ folded time sequences, so its cost grows with the channel count, not with batch $\times$ channels, while the causal time scan is enabled only where the channel count keeps it affordable. A wall-clock, memory, and throughput comparison against S-Mamba, iTransformer, and PatchTST under identical hardware is left for the extended version.

5 Conclusion

DD-Mamba shows that a state-space forecaster benefits from committing to *both* domains at once — provided each branch carries a linear pathway of its domain, deep components are zero-initialized corrections, and capacity (especially cross-variate mixing) is placed according to dataset characteristics rather than uniformly. Under the unified look-back-96 protocol, this recipe achieves the lowest reported average MSE on seven of nine standard benchmarks and remains competitive on Exchange, while still lagging behind S-Mamba on Traffic, with parameter counts spanning 0.09M to 19.7M across datasets. Two-dataset, partly multi-seed ablations validate the variate mixer consistently, find the linear anchor, time branch, and temporal SSM helpful on some datasets and neutral on others, and surface a transferable methodological caution: instance normalization can help or hurt substantially depending on whether apparent nonstationarity is real shift or periodic structure.

Limitations remain. Methodologically: the complete ablation suite covers two datasets at one horizon (one of them multi-seed), and baseline numbers are quoted from [12] under an identical protocol rather than re-run in a unified framework — our in-framework DLinear re-run on ETTh1 (Sect. 4.1) is a first step, and reproducing the remaining baselines is planned — so our comparative claims should be read as “competitive with reported results under the same protocol”. Architecturally: the fixed look-back of 96 hides cycles longer than the window (weather’s daily and traffic’s weekly periods); we hypothesize this explains the remaining deficit on traffic, a claim to be tested with longer look-backs (168/336), calendar covariates, and a same-window re-comparison against S-Mamba. Multi-resolution windows and adaptive spectral cutoffs are natural next steps.

Reproducibility. Code, configurations, and the ablation harness are provided as supplementary material and will be released in a public repository upon acceptance.

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